

Lab 5: Securing Apache Web Server

Introduction

This lab focuses on implementing secure web communication using HTTPS (HTTP Secure) protocol. The HTTP protocol, while widely used, transmits data in plaintext, making it vulnerable to eavesdropping and man-in-the-middle attacks. To address these security concerns, SSL/TLS (Secure Sockets Layer/Transport Layer Security) was introduced to provide encryption, authentication, and data integrity.

HTTPS provides three critical security features:

1. Confidentiality: Encrypts data between client and server
2. Integrity: Prevents data tampering during transmission
3. Authentication: Verifies server identity through certificates

Setup and Environment

```
This message is shown once a day. To disable it please create the
/home/jerryberry/.hushlogin file.
jerryberry@DESKTOP-DM05L5A:/mnt/c/WINDOWS/system32$ sudo apt update
[sudo] password for jerryberry:
```

Task 1: Became a Certificate Authority

```
Processing triggers for libc-bin (2.35-0ubuntu0.0) ...
jerryberry@DESKTOP-DM05L5A:/mnt/c/WINDOWS/system32$ cd ins-lab
jerryberry@DESKTOP-DM05L5A:/mnt/c/WINDOWS/system32/ins-lab$ mkdir lab-5
jerryberry@DESKTOP-DM05L5A:/mnt/c/WINDOWS/system32/ins-lab$ cd lab-5
jerryberry@DESKTOP-DM05L5A:/mnt/c/WINDOWS/system32/ins-lab/lab-5$ cp /usr/lib/ssl/openssl.cnf .
jerryberry@DESKTOP-DM05L5A:/mnt/c/WINDOWS/system32/ins-lab/lab-5$ mkdir -p demoCA/certs
mkdir -p demoCA/crl
mkdir -p demoCA/newcerts
mkdir -p demoCA/private
jerryberry@DESKTOP-DM05L5A:/mnt/c/WINDOWS/system32/ins-lab/lab-5$ touch demoCA/index.txt
echo "1000" > demoCA/serial
jerryberry@DESKTOP-DM05L5A:/mnt/c/WINDOWS/system32/ins-lab/lab-5$ nano openssl.cnf
```

Generating self-signed CA Certificate, The output of the command are stored in two files: ca.key and ca.crt

```
jerryberry@DESKTOP-DM05L5A:/mnt/c/WINDOWS/system32/ins-lab/lab-5$ openssl req -new -x509 -keyout ca.key -out ca.crt -config openssl.cnf -days 365
+++++
.....
+++++
Enter PEM pass phrase:
Verifying - Enter PEM pass phrase:
-----
You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
```

Generated public/private key pair

```
Email Address []:jerinusrat001@gmail.com
jerryberry@DESKTOP-DMO5LSA:/mnt/c/WINDOWS/system32/ins-lab/lab-5$ openssl genrsa -out example.com.key 2048
```

Generated a Certificate Signing Request (CSR)

```
jerryberry@DESKTOP-DMO5LSA:/mnt/c/WINDOWS/system32/ins-lab/lab-5$ openssl req -new -key example.com.key -out example.com-new.csr -config openssl.cnf
You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value.
```

Task2: Signing certificate with my CA

Signed the certificate with my CA for example.com

```
A challenge password []:
An optional company name []:
jerryberry@DESKTOP-DMO5LSA:/mnt/c/WINDOWS/system32/ins-lab/lab-5$ openssl ca -in example.com.csr -out example.com.crt -cert ca.crt -keyfile ca.key -config openssl.cnf -days 365
Using configuration from openssl.cnf
Enter pass phrase for ca.key:
Check that the request matches the signature
Signature ok
The stateOrProvinceName field is different between
CA certificate (Chittagong) and the request (Dhaka)
```

```
CA certificate (Chittagong) and the request (Dhaka)
jerryberry@DESKTOP-DMO5LSA:/mnt/c/WINDOWS/system32/ins-lab/lab-5$ openssl ca -in example.com-new.csr -out example.com.crt -cert ca.crt -keyfile ca.key -config openssl.cnf -days 365
Using configuration from openssl.cnf
Enter pass phrase for ca.key:
```

```
An optional company name []:
jerryberry@DESKTOP-DMO5LSA:/mnt/c/WINDOWS/system32/ins-lab/lab-5$ openssl ca -in example.com-final.csr -out example.com.crt -cert ca.crt -keyfile ca.key -config openssl.cnf -days 365
Using configuration from openssl.cnf
Enter pass phrase for ca.key:
Check that the request matches the signature
Signature ok
Certificate Details:
    Serial Number: 4096 (0x1000)
    Validity
        Not Before: Nov  8 13:28:08 2025 GMT
        Not After : Nov  8 13:28:08 2026 GMT
    Subject:
        countryName             = BD
        stateOrProvinceName     = Chittagong
        organizationName        = INS-Lab
        organizationalUnitName  = IT dept
        commonName               = example.com
        emailAddress             = jerinusrat001@gmail.com
    X509v3 extensions:
        X509v3 Basic Constraints:
            CA:FALSE
        X509v3 Subject Key Identifier:
            8F:A2:09:7E:42:C1:B2:B3:C6:51:98:EB:AD:DB:02:F2:AF:DB:7E:54
        X509v3 Authority Key Identifier:
```

Important Notification
Allow important notifications
You will get important notifications
Tool when do not disturb

```
EA:8C:20:88:A8:69:0A:4E:35:C0:F1:5B:61:D9:4A:E8:53:AE:36:8D
Certificate is to be certified until Nov  8 13:28:08 2026 GMT (365 days)
Sign the certificate? [y/n]:Y

1 out of 1 certificate requests certified, commit? [y/n]y
Write out database with 1 new entries
Database updated
```

Signed the certificate with my CA for webserverlab.com

```
jerryberry@DESKTOP-DMO5LSA:/mnt/c/WINDOWS/system32/ins-lab/lab-5$ openssl req -new -key webserverlab.com.key -out webserverlab.com-n.csr -config openssl.cnf
You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
Country Name (2 letter code) [AU]:BD
State or Province Name (full name) [Some-State]:Chittagong
Locality Name (eg, city) []:Chittagong
Organization Name (eg, company) [Internet Widgits Pty Ltd]:INS-Lab
Organizational Unit Name (eg, section) []:Dev
Common Name (e.g. server FQDN or YOUR name) []:webserverlab.com
Email Address []:jerinusrat001@gmail.com

Please enter the following 'extra' attributes
to be sent with your certificate request
A challenge password []:
An optional company name []:
```

```

jerryberry@DESKTOP-DM05L5A:/mnt/c/WINDOWS/system32/ins-lab/lab-5$ openssl ca -in webserverlab.com-n.csr -out webserverlab.com.crt -cert ca.crt -keyfile ca.key -config openssl.cnf -days 365
Using configuration from openssl.cnf
Enter pass phrase for ca.key:
Check that the request matches the signature
Signature ok
Certificate Details:
  Serial Number: 4097 (0x1001)
  Validity
    Not Before: Nov  8 13:48:20 2025 GMT
    Not After : Nov  8 13:48:20 2026 GMT

```

Signed the certificated and stored them

```

jerryberry@DESKTOP-DM05L5A:/mnt/c/WINDOWS/system32/ins-lab/lab-5$ ls -la *.crt *.key *.csr
-rwxrwxrwx 1 jerryberry jerryberry 1480 Nov  8 19:06 ca.crt
-rwxrwxrwx 1 jerryberry jerryberry 1854 Nov  8 19:04 ca.key
-rwxrwxrwx 1 jerryberry jerryberry 1074 Nov  8 19:27 example.com-final.csr
-rwxrwxrwx 1 jerryberry jerryberry 1078 Nov  8 19:23 example.com-new.csr
-rwxrwxrwx 1 jerryberry jerryberry 4553 Nov  8 19:28 example.com.crt
-rwxrwxrwx 1 jerryberry jerryberry 1066 Nov  8 19:18 example.com.csr
-rwxrwxrwx 1 jerryberry jerryberry 1704 Nov  8 19:14 example.com.key
-rwxrwxrwx 1 jerryberry jerryberry 1074 Nov  8 19:46 webserverlab.com-n.csr
-rwxrwxrwx 1 jerryberry jerryberry 4554 Nov  8 19:48 webserverlab.com.crt
-rwxrwxrwx 1 jerryberry jerryberry 1078 Nov  8 19:44 webserverlab.com.csr
-rwxrwxrwx 1 jerryberry jerryberry 1704 Nov  8 19:42 webserverlab.com.key
jerryberry@DESKTOP-DM05L5A:/mnt/c/WINDOWS/system32/ins-lab/lab-5$

```

TASK 3: DEPLOY HTTPS IN APACHE

Enabled Apache SSL Module

```

jerryberry@DESKTOP-DM05L5A:/mnt/c/WINDOWS/system32/ins-lab/lab-5$ sudo a2enmod ssl

```

Create Website Directories

```

systemctl restart apache2
jerryberry@DESKTOP-DM05L5A:/mnt/c/WINDOWS/system32/ins-lab/lab-5$ sudo systemctl restart apache2
jerryberry@DESKTOP-DM05L5A:/mnt/c/WINDOWS/system32/ins-lab/lab-5$ sudo mkdir -p /var/www/example.com/html
jerryberry@DESKTOP-DM05L5A:/mnt/c/WINDOWS/system32/ins-lab/lab-5$ sudo mkdir -p /var/www/webserverlab.com/html
jerryberry@DESKTOP-DM05L5A:/mnt/c/WINDOWS/system32/ins-lab/lab-5$ sudo bash -c 'echo "<html>
<head><title>Example.com Secure Site</title></head>
<body style=\"font-family: Arial, sans-serif; margin: 40px;\">
<h1>Salam to Example.com (HTTPS)</h1>
<p>This page is served securely via HTTPS</p>
<p><strong>Certificate issued by:</strong> INS Root IN</p>
<p>SSL/TLS Enabled and Working?</p>
<hr>
<p>INS-Lab 5: Securing Apache Web Server</p>
</body>
</html>" > /var/www/example.com/html/index.html'
jerryberry@DESKTOP-DM05L5A:/mnt/c/WINDOWS/system32/ins-lab/lab-5$ sudo bash -c 'echo "<html>
<head><title>WebServerLab.com Secure Site</title></head>
<body style=\"font-family: Arial, sans-serif; margin: 40px;\">
<h1>SALAM to WebServerLab.com (HTTPS)</h1>
<p>This page is served securely via HTTPS</p>
<p>INS Root IN</p>
<p><strong>SSL/TLS:</strong> Enabled and Working</p>
<hr>
<p>INS Lab 5: Securing Apache Web Server</p>
</body>
</html>" > /var/www/webserverlab.com/html/index.html'

```

Set proper permissions
Create SSL virtual host configuration
Copy pasted this :

```
<IfModule mod_ssl.c>
    <VirtualHost *:443>

        ServerAdmin admin@webserverlab.com
        ServerName webserverlab.com
        ServerAlias www.webserverlab.com
        DocumentRoot /var/www/webserverlab.com/html

        ErrorLog ${APACHE_LOG_DIR}/webserverlab.com_error.log
        CustomLog ${APACHE_LOG_DIR}/webserverlab.com_access.log combined

        SSLEngine on
        SSLCertificateFile /mnt/c/WINDOWS/system32/ins-lab/lab-5/webserverlab.com.crt
        SSLCertificateKeyFile /mnt/c/WINDOWS/system32/ins-lab/lab-5/webserverlab.com.key
        SSLCertificateChainFile /mnt/c/WINDOWS/system32/ins-lab/lab-5/ca.crt

        # Security headers
        Header always set Strict-Transport-Security "max-age=63072000; includeSubDomains; preload"
        Header always set X-Content-Type-Options nosniff
        Header always set X-Frame-Options DENY

    </VirtualHost>
</IfModule>
```

Enable SSL Sites using `sudo a2ensite *-ssl.conf`

Configured hosts file:

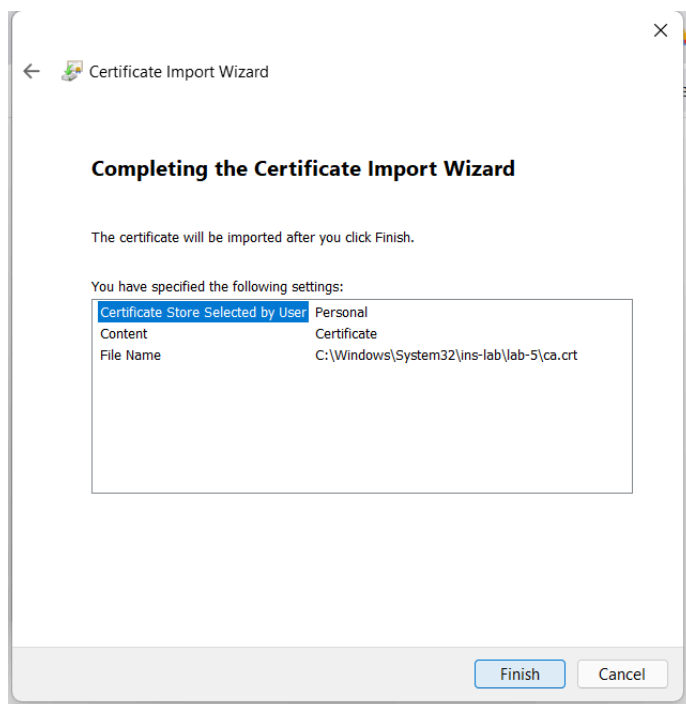
```
</html>
jerryberry@DESKTOP-DM05L5A:/mnt/c/WINDOWS/system32/ins-lab/lab-5$ sudo chown -R www-data:www-data /var/www/example.com
jerryberry@DESKTOP-DM05L5A:/mnt/c/WINDOWS/system32/ins-lab/lab-5$ sudo chown -R www-data:www-data /var/www/webserverlab.com
jerryberry@DESKTOP-DM05L5A:/mnt/c/WINDOWS/system32/ins-lab/lab-5$ sudo chmod -R 755 /var/www
jerryberry@DESKTOP-DM05L5A:/mnt/c/WINDOWS/system32/ins-lab/lab-5$ sudo nano /etc/apache2/sites-available/example.com-ssl.conf
jerryberry@DESKTOP-DM05L5A:/mnt/c/WINDOWS/system32/ins-lab/lab-5$ sudo nano /etc/apache2/sites-available/webserverlab.com-ssl.conf
jerryberry@DESKTOP-DM05L5A:/mnt/c/WINDOWS/system32/ins-lab/lab-5$ sudo a2ensite example.com-ssl.conf
Enabling site example.com-ssl.
To activate the new configuration, you need to run:
    systemctl reload apache2
jerryberry@DESKTOP-DM05L5A:/mnt/c/WINDOWS/system32/ins-lab/lab-5$ sudo a2ensite webserverlab.com-ssl.conf
Enabling site webserverlab.com-ssl.
To activate the new configuration, you need to run:
    systemctl reload apache2
jerryberry@DESKTOP-DM05L5A:/mnt/c/WINDOWS/system32/ins-lab/lab-5$ sudo apache2ctl configtest
AH00526: Syntax error on line 17 of /etc/apache2/sites-enabled/example.com-ssl.conf:
Invalid command 'Header', perhaps misspelled or defined by a module not included in the server configuration
jerryberry@DESKTOP-DM05L5A:/mnt/c/WINDOWS/system32/ins-lab/lab-5$ sudo nano /etc/apache2/sites-available/example.com-ssl.conf
jerryberry@DESKTOP-DM05L5A:/mnt/c/WINDOWS/system32/ins-lab/lab-5$ sudo apache2ctl configtest
AH00526: Syntax error on line 17 of /etc/apache2/sites-enabled/example.com-ssl.conf:
Invalid command 'Header', perhaps misspelled or defined by a module not included in the server configuration
jerryberry@DESKTOP-DM05L5A:/mnt/c/WINDOWS/system32/ins-lab/lab-5$ sudo nano /etc/apache2/sites-available/example.com-ssl.conf
jerryberry@DESKTOP-DM05L5A:/mnt/c/WINDOWS/system32/ins-lab/lab-5$ sudo nano /etc/apache2/sites-available/webserverlab.com-ssl.conf

</html>
jerryberry@DESKTOP-DM05L5A:/mnt/c/WINDOWS/system32/ins-lab/lab-5$ cat /etc/apache2/sites-available/example.com-ssl.conf | grep -i servername
cat /etc/apache2/sites-available/webserverlab.com-ssl.conf | grep -i servername
    ServerName example.com
    ServerName webserverlab.com
jerryberry@DESKTOP-DM05L5A:/mnt/c/WINDOWS/system32/ins-lab/lab-5$ echo "Example.com title:"
curl -s -k https://example.com | grep -o "<title>.*</title>"
Example.com title:
<title>Example.com Secure Site</title>
jerryberry@DESKTOP-DM05L5A:/mnt/c/WINDOWS/system32/ins-lab/lab-5$ echo "Webserverlab.com title:"
curl -s -k https://webserverlab.com | grep -o "<title>.*</title>"
Webserverlab.com title:
<title>WebServerLab.com Secure Site</title>
```

Deployed successfully

```
</body></html>
jerryberry@DESKTOP-DM05L5A:/mnt/c/WINDOWS/system32/ins-lab/lab-5$ curl -k -H "Host: example.com" https://172.17.158.10
<html>
<head><title>Example.com Secure Site</title></head>
<body style="font-family: Arial, sans-serif; margin: 40px;">
<h1>Salam to Example.com (HTTPS)</h1>
<p>This page is served securely via HTTPS</p>
<p><strong>Certificate issued by:</strong> INS Root IN</p>
<p>SSL/TLS Enabled and Working?</p>
<hr>
<p>INS-Lab 5: Securing Apache Web Server</p>
</body>
</html>
jerryberry@DESKTOP-DM05L5A:/mnt/c/WINDOWS/system32/ins-lab/lab-5$ curl -k https://webserverlab.com
<html>
<head><title>WebServerLab.com Secure Site</title></head>
<body style="font-family: Arial, sans-serif; margin: 40px;">
<h1>SAlam to WebServerLab.com (HTTPS)</h1>
<p>This page is served securely via HTTPS</p>
<p> INS Root IN</p>
<p><strong>SSL/TLS:</strong> Enabled and Working</p>
<hr>
<p>INS Lab 5: Securing Apache Web Server</p>
</body>
</html>
```

- Added my newly created certificate to Brave browser
- Cleared all cache memory



Task 4: HTTPS is working

WebBrowser View for [example.com](https://172.17.158.10)

<

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🔄

🔒

Not secure

https://172.17.158.10

Salam to Example.com (HTTPS)

This page is served securely via HTTPS

Certificate issued by: INS Root IN

SSL/TLS Enabled and Working?

INs-Lab 5: Securing Apache Web Server

Certificate :

Certificate Viewer: example.com

General

Details

Issued To

Common Name (CN)

example.com

Organisation (O)

INS-Lab

Organisational Unit (OU)

IT dept

Issued By

Common Name (CN)

INS Root IN

Organisation (O)

INS-Lab

Organisational Unit (OU)

Lab-Task5

Validity Period

Issued On

Saturday 8 November 2025 at 19:28:08

Expires On

Sunday 8 November 2026 at 19:28:08

SHA-256 Fingerprints

Certificate

bc65f807b13e30f8114f2915023677e983f60cda99264569699f96ad41eb665a

Public key

7d25e917c4f52b886e1e84de63c0e5e7675ea5616355de76c70aa3f2aa7ce9f9

WebBrowser View for example.com

<

>

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⚠️ Not secure https://webserverlab.com

SAlam to WebServerLab.com (HTTPS)

This page is served securely via HTTPS

INS Root IN

SSL/TLS: Enabled and Working

INS Lab 5: Securing Apache Web Server

Certificate:

Certificate Viewer: webserverlab.com

General

Details

Issued To

Common Name (CN)

webserverlab.com

Organisation (O)

INS-Lab

Organisational Unit (OU)

Dev

Issued By

Common Name (CN)

INS Root IN

Organisation (O)

INS-Lab

Organisational Unit (OU)

Lab-Task5

Validity Period

Issued On

Saturday 8 November 2025 at 19:48:20

Expires On

Sunday 8 November 2026 at 19:48:20

SHA-256 Fingerprints

Certificate

e4a31cce2cc522f1af0981483b90b7a2749445dd2097a6cacbfef03936b8ce6

Public key

640dc8f667d3cdce47c70f22c0c940cf2d01dba7f86a2c9de33c1c3835t2152

- Created my own Certificate Authority (INS Root IN)
- Generated SSL certificates for both example.com and webserverlab.com
- Configured Apache with SSL virtual hosts on port 443
- Both sites now serve content securely via HTTPS
- The certificates are trusted because I imported my CA certificate

```
jerryberry@DESKTOP-DM05L5A:/mnt/c/WINDOWS/system32/ins-lab/lab-5$ sudo apache2ctl -S
[sudo] password for jerryberry:
Sorry, try again.
[sudo] password for jerryberry:
VirtualHost configuration:
*:80      DESKTOP-DM05L5A.localdomain (/etc/apache2/sites-enabled/000-default.conf:1)
*:443    is a NameVirtualHost
          default server example.com (/etc/apache2/sites-enabled/example.com-ssl.conf:2)
          port 443 namevhost example.com (/etc/apache2/sites-enabled/example.com-ssl.conf:2)
              alias www.example.com
          port 443 namevhost webserverlab.com (/etc/apache2/sites-enabled/webserverlab.com-ssl.conf:2)
              alias www.webserverlab.com
ServerRoot: "/etc/apache2"
Main DocumentRoot: "/var/www/html"
Main ErrorLog: "/var/log/apache2/error.log"
Mutex watchdog-callback: using_defaults
Mutex ssl-stapling-refresh: using_defaults
Mutex ssl-stapling: using_defaults
Mutex ssl-cache: using_defaults
Mutex default: dir="/var/run/apache2/" mechanism=default
PidFile: "/var/run/apache2/apache2.pid"
Define: DUMP_VHOSTS
Define: DUMP_RUN_CFG
User: name="www-data" id=33
Group: name="www-data" id=33
```